

LOW STYRENE SBR PATENT LANDSCAPE: SOLUTION AND EMULSION ROUTES, FEED CONTROL, AND FUNCTIONALIZED COPOLYMER VARIANTS

1. ABSTRACT

This report synthesizes four primary patents centered on styrene-butadiene rubber (SBR) and styrene-butadiene copolymer manufacture, with emphasis on low-styrene or reduced-styrene embodiments relevant to low styrene SBR. The selected patents span solution polymerization, continuous two-zone feed control, barium-based catalyst systems, and post-polymerization functionalization of solution SBR. Across the set, the target material is consistently a butadiene-rich elastomeric copolymer, but the disclosed styrene contents vary by process and embodiment, ranging from about 10 wt% in a targeted low-vinyl SSBR example to 20-30 wt% in a tapered low-molecular-weight copolymer and 10/90 styrene/butadiene feed compositions that yield random SBR. The landscape shows that styrene reduction is not always the sole objective; instead, patents often optimize microstructure, randomness, molecular weight, and terminal functionality while maintaining tire-relevant properties such as storage modulus, wear resistance, and low vinyl content.

The major polymerization approaches are solution-based. US6372863B1 teaches continuous two-reactor solution polymerization with split-feed control to suppress tapered block formation and achieve random SBR with low vinyl content. US6103842A discloses solution polymerization in hydrocarbon solvent using a barium alkoxide/lithium alkoxide/organolithium catalyst system to produce highly random SBR with low vinyl and high trans character. US11591428B2 focuses on low molecular weight tapered styrene-butadiene copolymers prepared in hexane with organolithium initiation and sodium dodecylbenzenesulfonate, then isolated by coagulation and drying. US10174133B2 describes modified SSBR prepared in hexane with n-butyllithium, polar additive, terminal butadiene capping, and silane-containing modifier, followed by ethanol termination and water-assisted solvent removal. Overall, the patents show a strong process-control trend: monomer feed ratio, staged addition, catalyst/modifier selection, and downstream isolation are used to tune styrene distribution and performance rather than merely lowering styrene content alone.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US11591428B2
- Patent Title: Synthesis of low molecular weight tapered styrene-butadiene copolymer and its ...
- Assignee: Bridgestone Corporation
- Jurisdiction: US
- Publication Year: 2023
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses solution polymerization of styrene and 1,3-butadiene in hydrocarbon solvent to make low molecular weight tapered styrene-butadiene copolymers. The examples use hexane, organolithium initiation, and sodium dodecylbenzenesulfonate, followed by coagulation and drum drying to isolate the polymer.

- Hexane: 0.18 kg — Example 1 - Styrene solution concentration: 31.8 wt % in hexane — Example 1 - 1,3-butadiene solution concentration: 20.2 wt % in hexane — Example 1 - SDBS: 2.27 mL of 0.5 M in ethylbenzene — Example 1 - n-butyl lithium: 6.05 mL of 1.60 M in hexane — Example 1 - Reaction temperature: 60° C. — Example 1 - Batch temperature peak: 69° C. — Example 1 - Post-reaction cooling temperature: 37° C. — Example 1 - Mn: 52 kg/mol — Example 1 - Mw: 55 kg/mol — Example 1 - PDI: 1.05 — Example 1 - Tg: -76.07° C. — Example 1 - Percent styrene: 26.6% — Example 1 - Styrene sequences >3 units: 38.7% of styrene units — Example 1 - Styrene sequences >10 units: 10.9% of styrene units — Example 1 - Vinyl: 9.4% — Example 1 - 1,4-butadiene incorporation: 64% — Example 1 - Cis-microstructure in butadiene units: 39.4 weight percent — Example 1 - Trans-microstructure in butadiene units: 49.6 weight percent — Example 1 - Hexane: 0.30 kg — Example 2 - Styrene solution concentration: 32.9 wt % in hexane — Example 2 - 1,3-butadiene solution concentration: 20 wt % in hexane — Example 2 - SDBS: 3.63 mL of 0.5 M in ethylbenzene — Example 2 - n-butyl lithium: 9.07 mL of 1.60 M in hexane — Example 2 - Reaction temperature: 66° C. — Example 2 - Batch temperature peak: 107° C. — Example 2 - Post-reaction cooling temperature: 46° C. — Example 2 - Mn: 66 kg/mol — Example 2 - Mw: 74 kg/mol — Example 2 - PDI: 1.12 — Example 2 - Tg: -78.79° C. — Example 2 - Percent styrene: 25.7% — Example 2 - Styrene sequences >3 units: 48.6% of styrene units — Example 2 - Styrene sequences >10 units: 17.5% of styrene units — Example 2 - Vinyl: 9.7% — Example 2 - 1,4-butadiene incorporation: 64% — Example 2 - Cis-microstructure in butadiene units: 38.5 weight percent — Example 2 - Trans-microstructure in butadiene units: 50.1 weight percent — Example 2 - Medium & Water Role: Organic hydrocarbon solution polymerization in hexane; water is not part of the polymerization medium. Water-related processing is not disclosed in the extracted evidence, while isolation is by coagulation in isopropanol and drum drying. - styrene

content below conventional SBR ranges: 25.7%

Medium & Water Role - Organic hydrocarbon solution polymerization in hexane; water is not part of the polymerization medium. Water-related processing is not disclosed in the extracted evidence, while isolation is by coagulation in isopropanol and drum drying. - Core reaction medium: Organic hydrocarbon solution polymerization in hexane

styrene content below conventional SBR ranges - 25.7%

Relevant Experimental Evidence - Example 1: Produced a tapered styrene-butadiene copolymer with 26.6% styrene, low vinyl content, and Mn/Mw near 52/55 kg/mol. - Example 1: Demonstrated improved wear resistance when used as an oil replacement in carbon-black-filled rubber compositions. - Example 2: Produced a similar tapered copolymer with 25.7% styrene and Mn/Mw near 66/74 kg/mol. - Example 2: Showed the same material class can be tuned by feed and temperature to maintain low vinyl and targeted Tg. - Example 1-2: Both examples isolated the cement by coagulation in isopropanol containing BHT and drum drying.

Technical Relevance This patent is relevant because it targets styrene-butadiene copolymers with controlled styrene distribution and relatively low molecular weight, which are directly applicable to low styrene SBR design. Although the disclosed styrene levels are not ultra-low, the process and property control are central to tuning butadiene-rich SBR grades for tire performance.

Patent 2

- Patent Number: US6372863B1
- Patent Title: Synthesis of styrene-butadiene rubber
- Assignee: The Goodyear Tire & Rubber Company
- Jurisdiction: US
- Publication Year: 2002
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses continuous solution polymerization of styrene and 1,3-butadiene in hydrocarbon solvent using anionic chain polymerization, with split-feed control across two reactors to suppress tapered block formation. The process is designed to produce random SBR with low vinyl content and controlled styrene sequence distribution. - Styrene target: 18 wt % — Table I - Vinyl target: 10 wt % — Table I - Mooney viscosity target: 40+/-5 ML1+4 (100° C.) — Table I - Styrene target: 25 wt % — Table I - Styrene target: 30 wt % — Table I - Mooney viscosity target: 45+/-5 OE (~110 base) — Table I - Initiator: n-butyllithium — Examples - Agitation speed: 200-250 rpm — Examples - Termination agent: rosin acid at 1 phr — Examples - Antioxidant: Wingstay K at 0.5 phr — Examples - Oil addition: 37.5 parts — Sample Type 4 — Examples - 1,2-butadiene added for gel suppression: 100-150 ppm — Preparations - Monomer solution concentration: 14 to 15 weight percent in

hexanes — Preparations - Polymerization temperature: 70° C. to 100° C. — Abstract/Claims - First-zone conversion: 60 percent to 90 percent — Claims/Description - Second-zone 1,3-butadiene split: 20 percent to 40 percent of total 1,3-butadiene — Claims/Description - Second-zone 1,3-butadiene conversion: at least 90 percent — Claims/Description - Second-zone total conversion limit: maximum 95 percent — Claims/Description - First-zone styrene charge: at least 5 percent more than total styrene bound — Description - First-zone styrene charge preferred: at least 7 percent more — Description - First-zone styrene charge more preferred: at least 10 percent more — Description - Polymerization medium monomer content: 5 to 30 weight percent — Description - Polymerization medium monomer content preferred: 10 to 25 weight percent — Description - Polymerization medium monomer content more preferred: 15 to 20 weight percent — Description - Styrene content in SBR: about 5 weight percent to less than 30 weight percent — Description - Less than 5 percent of styrene repeat units in blocks containing five or more units — Description - Less than 2 percent of styrene repeat units in blocks containing five or more units for less than about 30 wt% bound styrene — Description - Medium & Water Role: Organic hydrocarbon solution polymerization in hexanes; water is not the polymerization medium. Water appears later in downstream isolation as steam stripping and extruder dewatering for some samples. - styrene content below conventional SBR ranges: 18 wt%

Medium & Water Role - Organic hydrocarbon solution polymerization in hexanes; water is not the polymerization medium. Water appears later in downstream isolation as steam stripping and extruder dewatering for some samples. - Core reaction medium: Organic hydrocarbon solution polymerization in hexanes

styrene content below conventional SBR ranges - 18 wt%

Relevant Experimental Evidence - Table I: Defined target polymers with styrene levels of 18, 25, and 30 wt% and low vinyl target of 10 wt%. - Examples: Used n-butyllithium in continuous two-reactor solution polymerization with 200-250 rpm agitation. - Examples: Terminated living chains with rosin acid and added antioxidant before isolation. - Preparations: Monomer solutions were made in hexanes at 14-15 wt% with 1,2-butadiene added for gel suppression. - Examples: Reported a broad set of solution SBR compositions from 10/90 to 80/20 styrene/butadiene with low vinyl content and random styrene sequences.

Technical Relevance This patent is highly relevant because it directly addresses random solution SBR synthesis with controlled styrene incorporation, low vinyl content, and feed strategies that avoid tapered block formation. It provides a process framework for butadiene-rich SBR grades, including low-styrene embodiments such as 18 wt% styrene.

Patent 3

- Patent Number: US6103842A
- Patent Title: Process and catalyst system for synthesizing random trans SBR with low vinyl ...
- Assignee: The Goodyear Tire & Rubber Company
- Jurisdiction: US

- Publication Year: 2000
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses solution polymerization of styrene and 1,3-butadiene in an organic solvent using a catalyst system composed of an organolithium compound, a barium alkoxide, and a lithium alkoxide. The process is intended to produce random SBR with low vinyl content and high trans character under isothermal conditions. - Premix composition: 20 weight percent 1,3-butadiene monomer in hexanes — Homopolymerization example - Barium 2-ethyl hexoxide: 0.80 mmole — Homopolymerization example - n-butyllithium: 1.6 mmole — Homopolymerization example - Lithium tertiary butoxide: 6.40 mmole — Homopolymerization example - Reaction temperature: 70° C. — Homopolymerization example - Polymerization time: 3 hours — Homopolymerization example - Styrene/butadiene premix composition: 10 percent/90 percent — Preparation of Solution SBR - Barium 2-ethyl hexoxide: 1 mmole per 100 g of monomer — Preparation of Solution SBR - n-butyl lithium: 2 mmole per 100 g of monomer — Preparation of Solution SBR - Lithium t-butoxide: 4 mmole per 100 g of monomer — Preparation of Solution SBR - Premix temperature: 80° C. — Preparation of Solution SBR - Polymer composition: 10 percent styrene and 90 percent butadiene — Preparation of Solution SBR - Polymer composition: 15 percent styrene and 85 percent butadiene — Table I - Polymer composition: 20 percent styrene and 80 percent butadiene — Table I - Polymer composition: 25 percent styrene and 75 percent butadiene — Table I - Polymer composition: 30 percent styrene and 70 percent butadiene — Table I - Polymer composition: 35 percent styrene and 65 percent butadiene — Table I - Polymer composition: 40 percent styrene and 60 percent butadiene — Table I - Polymer composition: 50 percent styrene and 50 percent butadiene — Table I - Polymer composition: 60 percent styrene and 40 percent butadiene — Table I - Polymer composition: 70 percent styrene and 30 percent butadiene — Table I - Polymer composition: 80 percent styrene and 20 percent butadiene — Table I - Tg: -84° C. — Table I, 10/90 polymer - 1,2 content: 8% — Table I, 10/90 polymer - Mooney: 61.0 — Table I, 10/90 polymer - Tg: -78° C. — Table I, 15/85 polymer - 1,2 content: 8% — Table I, 15/85 polymer - Mooney: 62 — Table I, 15/85 polymer - Tg: -73° C. — Table I, 20/80 polymer - 1,2 content: 7% — Table I, 20/80 polymer - Mooney: 64 — Table I, 20/80 polymer - Tg: -66° C. — Table I, 25/75 polymer - 1,2 content: 7% — Table I, 25/75 polymer - Mooney: 44 — Table I, 25/75 polymer - Tg: -61° C. — Table I, 30/70 polymer - 1,2 content: 6% — Table I, 30/70 polymer - Mooney: 40 — Table I, 30/70 polymer - Tg: -53° C. — Table I, 35/65 polymer - 1,2 content: 6% — Table I, 35/65 polymer - Mooney: 39 — Table I, 35/65 polymer - Tg: -48° C. — Table I, 40/60 polymer - 1,2 content: 6% — Table I, 40/60 polymer - Mooney: 32 — Table I, 40/60 polymer - Tg: -29° C. — Table I, 50/50 polymer - 1,2 content: 6% — Table I, 50/50 polymer - Mooney: 34 — Table I, 50/50 polymer - Tg: -14° C. — Table I, 60/40 polymer - 1,2 content: 6% — Table I, 60/40 polymer - Mooney: 31 — Table I, 60/40 polymer - Tg: 11° C. — Table I, 70/30 polymer - 1,2 content: not reported — Table I, 70/30 polymer - Mooney: 39 — Table I, 70/30 polymer - Tg: 35° C. — Table I, 80/20 polymer - Medium & Water Role: Organic hydrocarbon solution polymerization in hexanes; no water-related process step is disclosed in the extracted

evidence. The polymerization and isolation are described entirely in organic solvent under nitrogen. - styrene content below conventional SBR ranges: 10 percent styrene

Medium & Water Role - Organic hydrocarbon solution polymerization in hexanes; no water-related process step is disclosed in the extracted evidence. The polymerization and isolation are described entirely in organic solvent under nitrogen. - Core reaction medium: Organic hydrocarbon solution polymerization in hexanes

styrene content below conventional SBR ranges - 10 percent styrene

Relevant Experimental Evidence - Preparation of Solution SBR: A 10/90 styrene/butadiene premix gave a random copolymer with constant 10% styrene and 90% butadiene. - Table I: Demonstrated a composition series from 10/90 to 80/20 styrene/butadiene with corresponding Tg and Mooney values. - Examples: All SBR copolymers were reported to have low vinyl content and random styrene sequences. - Homopolymerization example: Showed the catalyst system also polymerizes butadiene alone at 70° C. for 3 hours. - Preparation of Solution SBR: Catalyst loading was specified as 1 mmole barium alkoxide per 100 g monomer, with 2 mmole n-butyl lithium and 4 mmole lithium t-butoxide per 100 g monomer.

Technical Relevance This patent is relevant because it provides a catalyst system and solution polymerization method for making random SBR with low vinyl content and tunable styrene/butadiene ratios. The disclosed 10/90 and 15/85 compositions are directly aligned with low-styrene SBR targets.

Patent 4

- Patent Number: US10174133B2
- Patent Title: ... styrene-butadiene copolymer, preparation method thereof, and rubber composition ...
- Assignee: Lg Chem, Ltd.
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses solution polymerization of styrene and 1,3-butadiene in normal hexane using n-butyllithium and a polar additive, followed by terminal capping with additional butadiene and post-polymerization modification with a silane-containing modifier. The polymer is terminated with ethanol and isolated by addition to warm water, solvent removal, and roll drying. - Styrene: 29.7 g — Example 1 - 1,3-butadiene: 78.1 g — Example 1 - Normal hexane: 550 g — Example 1 - 2,2-di(2-tetrahydrofuryl)propane (DTP): 0.124 g — Example 1 - Reaction temperature: 50° C. — Example 1 - n-butyllithium: 3 mmol — Example 1 - Additional 1,3-butadiene for capping: 2.2 g —

Example 1 - Modifier: 0.181 g of 2-methoxyethyl 3-(bis(3-triethoxysilyl)propyl)amino)propanoate — Example 1 - [DTP]/[act. Li]: 2.3 — Example 1 - [modifier]/[act. Li]: 1.0 — Example 1 - Styrene: 270 g — Example 2 - 1,3-butadiene: 710 g — Example 2 - Normal hexane: 5 kg — Example 2 - 2,2-di(2-tetrahydrofuryl)propane (DTP): 0.94 g — Example 2 - Reaction temperature: 40° C. — Example 2 - n-butyllithium: 25.40 g (2.62 wt % in hexane, 33% activation) — Example 2 - Additional 1,3-butadiene for capping: 20.0 g — Example 2 - Modifier: 1.98 g of ethyl 3-(bis(3-triethoxy(methyl)silyl)propyl)amino)propanoate — Example 2 - [DTP]/[act. Li]: 1.46 — Example 2 - [modifier]/[act. Li]: 0.85 — Example 2 - Ethanol: used to stop polymerization — Example 1 and Example 2 - BHT solution: 0.3 wt % in hexane — Example 1 - Wingstay K solution: 30 wt % in hexane — Example 2 - Medium & Water Role: Organic solution polymerization in normal hexane; water is not the polymerization medium. Water is used later for post-treatment, where the polymer is added to warm water heated using steam to remove solvents before roll drying. - styrene content below conventional SBR ranges: Not disclosed in extracted evidence

Medium & Water Role - Organic solution polymerization in normal hexane; water is not the polymerization medium. Water is used later for post-treatment, where the polymer is added to warm water heated using steam to remove solvents before roll drying. - Core reaction medium: Organic solution polymerization in normal hexane

styrene content below conventional SBR ranges - Not disclosed in extracted evidence

Relevant Experimental Evidence - Example 1: Prepared a modified SSBR at 50° C. with DTP, n-butyllithium, butadiene capping, and silane-containing modifier. - Example 1: Polymer was terminated with ethanol and isolated by warm-water solvent removal and roll drying. - Example 2: Repeated the process at 40° C. on a larger scale with different modifier and DTP/[act. Li] ratio. - Experimental Example 1: Showed improved combination efficiency and molecular weight metrics for the modified copolymers. - Comparative Example: Demonstrated that changing the modifier altered the modification efficiency relative to the conventional modifier.

Technical Relevance This patent is relevant because it discloses solution SSBR synthesis and post-polymerization modification, including a butadiene-rich feed and process controls that affect polymer architecture. While not a low-styrene patent per se, it is directly within the SBR synthesis space and includes a composition with styrene/butadiene feed ratios suitable for low-styrene embodiments.

PATENT COMPARISON TABLE

Patent	Applicant	Priority	Medium & Water Role	styrene content below conventional SBR ranges	Key Findings

US11591428B2	Bridgestone Corporation	2023	Organic hydrocarbon solution polymerization in hexane; water is not part of the polymerization medium. Water-related processing is not disclosed in the extracted evidence, while isolation is by coagulation in isopropanol and drum drying.	25.7%	Example 1: Produced a tapered styrene-butadiene copolymer with 26.6% styrene, low vinyl content, and Mn/Mw near 52/55 kg
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US6372863B1	The Goodyear Tire & Rubber Company	2002	Organic hydrocarbon solution polymerization in hexanes; water is not the polymerization medium. Water appears later in downstream isolation as steam stripping and extruder dewatering for some samples.	18 wt%	Table I: Defined target polymers with styrene levels of 18, 25, and 30 wt% and low vinyl target of 10 wt%.
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US6103842A	The Goodyear Tire & Rubber Company	2000	Organic hydrocarbon solution polymerization in hexanes; no water-related process step is disclosed in the extracted evidence. The polymerization and isolation are described entirely in organic solvent under nitrogen.	10 percent styrene	Preparation of Solution SBR: A 10/90 styrene/butadiene premix gave a random copolymer with constant 10% styrene and 90%
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US10174133B2	Lg Chem, Ltd.	2019	Organic solution polymerization in normal hexane; water is not the polymerization medium. Water is used later for post-treatment, where the polymer is added to warm water heated using steam to remove solvents before roll drying.	Not disclosed in extracted evidence	Example 1: Prepared a modified SSBR at 50° C. with DTP, n-butyllithium, butadiene capping, and silane-containing modifie
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3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US6372863B1 and US6103842A both disclose solution SBR in hydrocarbon solvent, but US6372863B1 emphasizes split-feed continuous reactor control to suppress tapered block formation, whereas US6103842A emphasizes a barium alkoxide/lithium alkoxide/organolithium catalyst system to drive randomization and high trans character.
- US11591428B2 also uses solution polymerization, but its focus is low molecular weight tapered styrene-butadiene copolymer with styrene content around 25-27 wt%, higher than the 10 wt% example in US6103842A and the 18 wt% target in US6372863B1.
- US10174133B2 differs from the other three by adding a post-polymerization modification step with a silane-containing modifier and by using ethanol termination plus water-assisted solvent removal; the other patents primarily focus on polymerization recipe and reactor control.

- Styrene content spans from 10 wt%/10 percent styrene in US6103842A to 18 wt% in US6372863B1 and 25.7-26.6% in US11591428B2; US10174133B2 does not clearly disclose the final styrene content of the target polymer in the extracted evidence.
- All four patents use organic solvent-based solution processes rather than aqueous emulsion polymerization, but only US6372863B1 and US10174133B2 explicitly disclose downstream water-related steps in the extracted evidence.

4. CONCLUSION

Across the selected primary patents, low-styrene SBR is approached mainly through solution polymerization with careful control of monomer feed, catalyst system, and reactor staging rather than through a dedicated low-styrene claim alone. The most directly low-styrene disclosure is US6103842A, which reports a 10/90 styrene/butadiene solution SBR and a broader series up to 20/80 and beyond. US6372863B1 provides a continuous two-reactor split-feed process with a targeted 18 wt% styrene SSBR and low-vinyl/random microstructure control. US11591428B2 contributes a tapered low-molecular-weight copolymer platform with styrene around 25-27 wt% and performance benefits in rubber compounds. US10174133B2 adds a functionalized SSBR route with solution polymerization, terminal capping, and water-assisted isolation. Together, the patents show that low-styrene SBR development is tightly linked to solution-process architecture, monomer feed management, and downstream handling.

5. REFERENCES

- US11591428B2 | Synthesis of low molecular weight tapered styrene-butadiene copolymer and its ... | Bridgestone Corporation | US | 2023 | <https://patents.google.com/patent/US11591428B2/en>
- US6372863B1 | Synthesis of styrene-butadiene rubber | The Goodyear Tire & Rubber Company | US | 2002 | <https://patents.google.com/patent/US6372863B1/en>
- US6103842A | Process and catalyst system for synthesizing random trans SBR with low vinyl ... | The Goodyear Tire & Rubber Company | US | 2000 | <https://patents.google.com/patent/US6103842A/en>
- US10174133B2 | ... styrene-butadiene copolymer, preparation method thereof, and rubber composition ... | Lg Chem, Ltd. | US | 2019 | <https://patents.google.com/patent/US10174133B2/en>